

AI FOR SUSTAINABLE
DEVELOPMENT UNIT-3
BANANA PROBLEM

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```
PS C:\Users\Trishul Swamy HM\Downloads\SEM6\AISD\unit3_banana> python unit3_banana_analysis.py
```

```
Cross-validation RMSE mean: 541.38, std: 51.29
```

```
RMSE: 459.38
```

```
R2: 0.9466
```

```
Top 12 feature importances:
```

```
hour                0.825924
dayofweek            0.062383
is_weekend           0.048652
temp                0.032547
month                0.010771
clouds_all           0.005090
rain_1h              0.002650
weather_desc_overcast clouds 0.001034
weather_main_Clouds  0.001017
weather_desc_broken clouds 0.000970
weather_desc_sky is clear 0.000942
weather_desc_few clouds 0.000762
dtype: float64
```

```
Sample traffic predictions (first 10):
```

```
actual predicted
```

```
6364 6169.758333
```

```
2030 2178.516667
```

```
708 685.391667
```

```
2029 2013.750000
```

```
3734 4873.933333
```

```
4337 4427.591667
```

```
5821 5928.250000
```

```
5710 5620.341667
```

```
5017 4589.041667
```

```
3247 4159.950000
```

```
=== Synthetic Credit Score Model (Financial inclusion) ===
```

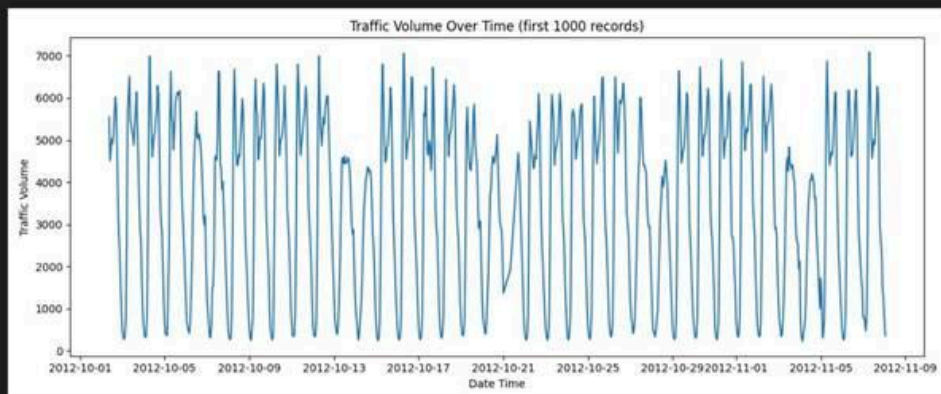
```
Accuracy: 1.0000
```

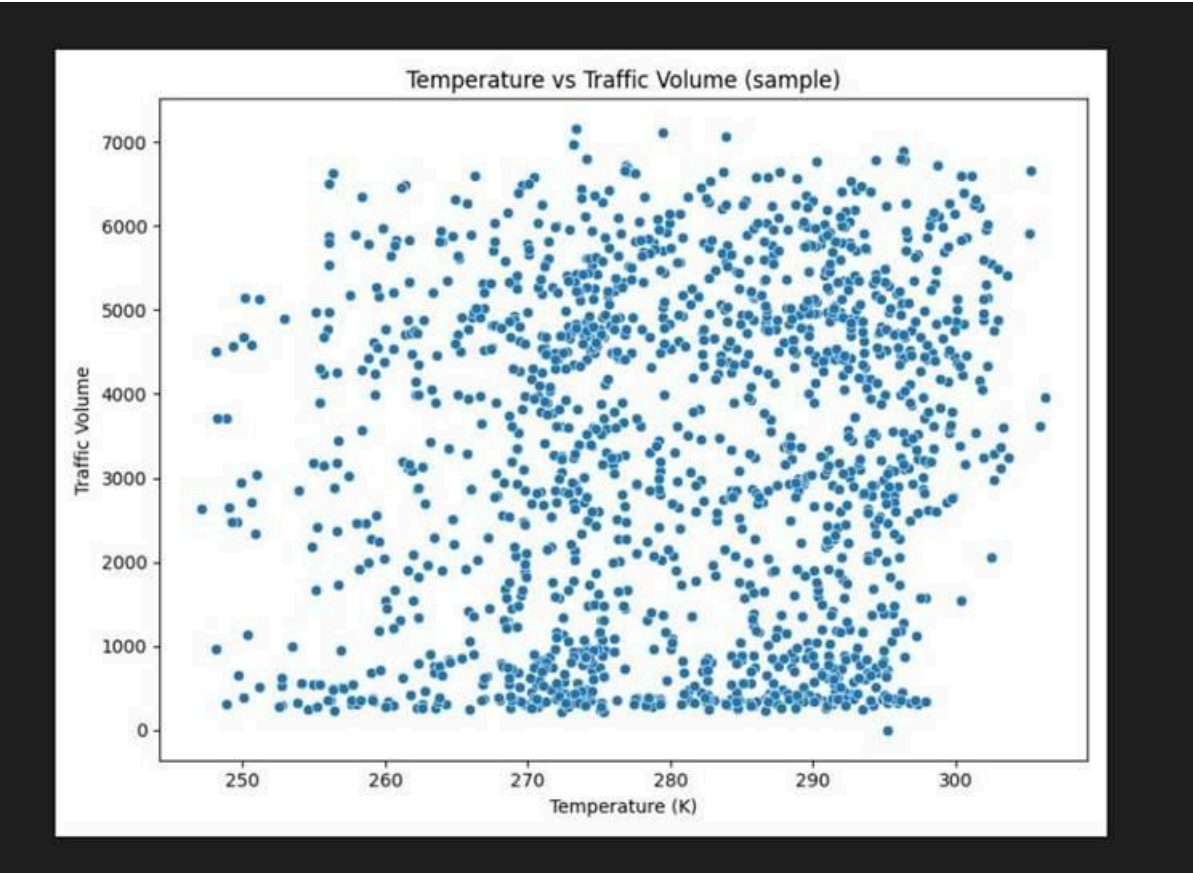
```
Credit classification report:
```

	precision	recall	f1-score	support
poor	1.0000	1.0000	1.0000	3000
accuracy			1.0000	3000
macro avg	1.0000	1.0000	1.0000	3000
weighted avg	1.0000	1.0000	1.0000	3000

```
Text report saved to C:\Users\Trishul Swamy HM\Downloads\SEM6\AISD\unit3_banana\unit3_banana_report_summary.txt
```

```
Markdown report saved to C:\Users\Trishul Swamy HM\Downloads\SEM6\AISD\unit3_banana\unit3_banana_report.md
```





PART2

df.head()

Out[5]:	Income	Age	LoanAmount	RepaymentHistory	Credi Utilization	CreditScore
0	36386	26	36985	10	0.35	Poor
1	93935	52	34417	6	0.48	Poor
2	92944	49	7454	6	0.50	Poor
3	90083	36	16851	8	0.73	Poor
4	22556	43	47000	4	0.38	Average

Accuracy: 1.0

Classification	Report: precision	recall	f1-score	support
0	1.00	1.00	1.00	553
1	1.00	1.00	1.00	395
2	1.00	1.00	1.00	52
accuracy			1.00	1000
macro avg	1.00	1.00	1.00	1000
weighted avg	1.00	1.00	1.00	1000

	Actual	Predicted
1501	1	1
2586	0	0
2653	1	1
1055	1	1
705	1	1
106	0	0
589	1	1
2468	1	1
2413	0	0
1600	0	0
2464	1	1
228	0	0
915	0	0
794	1	1
3021	0	0
3543	1	1
1073	1	1
3351	1	1
1744	0	0
1084	1	1

